Practice Sheet-03

(Exploratory Data Analysis)

1. For a bouquet, a florist measures the lengths of daisy petals and got the following 5 number summary: **Minimum: 1.00, Q1: 1.60, Median: 4.35, Q3: 5.10, Maximum: 6.90**
Analyze that data with the fence rule. Which statement is true?

Answers:

a) There are no outliers at all.
b) There is at least one low outliers, but no high outliers.
c) There are outliers, but it is unknown if they are on the low or high end.
d) There are no low outliers, but there is at least one high outlier.

2.The following is the age of members of a book group. Use the fence rule, are there any outliers?
Data: 16, 17, 20, 30, 43, 50, 51, 56, 62, 70, 76, 78, 85, 88, 91, 97

3.Given this five number summary of scores out of 20 on a quiz, determine if there are any outliers.
Minimum: 0.00, Q1: 1.10, Median: 4.35, Q3: 5.10, Maximum: 18.

4.In the line at the grocery store, here is the amount spent of the first 7 people:

$3, $9, $14, $25, $26, $85, $100. Based on the fence rule, which (if any) amount is an outlier?

5.Mr M saw 10 movies last year and recorded the length of each of them. The five number summary of his analysis is the following:

Minimum: 35, Q1: 58, Median: 75, Q3: 120, Maximum: 180

The movie he watched today was 200 minutes long. Compared to the five number summary, is 200 minutes an outlier? Apply the fence rule to decide.

6. At a hospital, babies are weighed in ounces. These are the weights of the newest 6 babies: **3.6, 5, 6, 6, 7.3, 10.** A baby born yesterday weighed 10.5 ounces. Using the Fence Rule, would that be considered an outlier?

7. The weather person displayed the five number summary of today's high temperatures from the past 80 years.
Minimum: 15, Q1: 45, Median: 75, Q3: 80, Maximum: 100
Use the Fence Rule, are there any outliers from this data and also construct the Boxplot?

8. Given below is the distribution of marks obtained by 140 students in an examination:

Marks	:	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
No. of Students	:	7	15	18	25	30	20	16	7	2

Find the outliers of the above distribution and construct the Boxplot.

9. Over the years, Jill has had several different dogs, each with varying weights. Altogether, the dog's weights have this five number summary: **Minimum: 5, Q1: 18, Median: 29, Q3: 31, Maximum: 55.** Construct the Boxplot and find if there is any outliers?

10. IILM campus conducted a campus placement drive for final-year B. Tech students. The Career Services team recorded the annual salary packages (in lakhs per annum) offered to 50 students. The salary packages are as follows:

3, 5, 6, 8, 7, 9, 10, 5, 12, 6, 8, 11, 4, 9, 13, 14, 6, 7, 8, 10, 3, 12, 5, 7, 9, 6, 11, 10, 8, 5, 9, 6, 14, 12, 7, 8, 13, 5, 9, 10, 4, 6, 7, 11, 10, 3, 12, 6, 8, 9.
Find if the above data contains an outlier and also construct the Boxplot.

11. Why is data standardization important in machine learning?

12. For a given dataset with the feature 10, 20, 30, 40, 50. Standardization of the data using z-score.

13. Standardize the dataset **X = [5, 15, 25, 35, 45, 55]** using Z-score normalization.